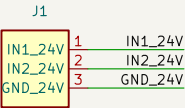
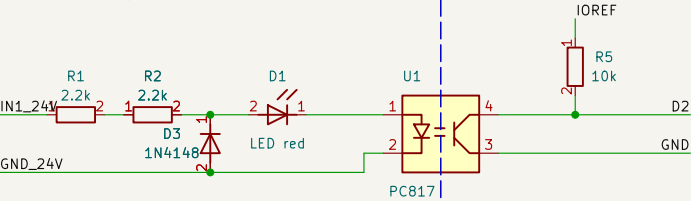


PC817 24 V input test bench – 2 channels – shield for Arduino UNO R4 WiFi / UNO Q

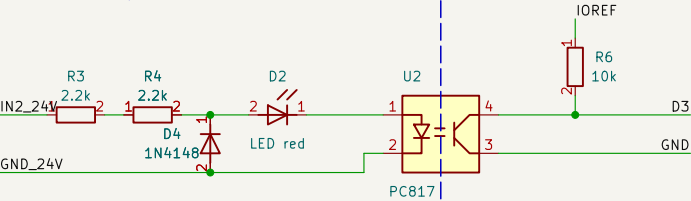
If = (24 V – 1.2 V opto – 2.0 V LED) / 4,4 k = 4.7 mA. Each 2.2 k (1/4 W); 49 mW at 24 V, 75 mW at 28.8 V.
1N4148 protects both LEDs against a reversed input (PC817 VR max = 6 V).
GND_24V and GND are NOT connected: this is the isolation barrier.
Pull-ups go to IOREF: 5 V on the UNO R4 WiFi, 3.3 V on the UNO Q. Never to the 5V pin. Sketch: pinMode(D2/D3, INPUT).

Channel 1: 24 V present -> D2 LOW, LED lit

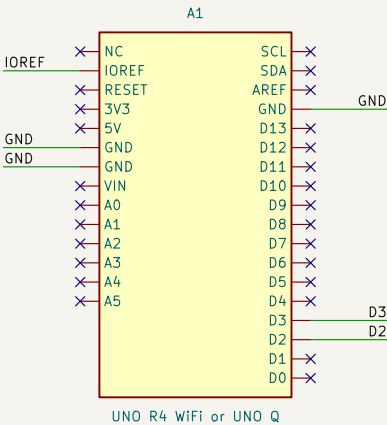


24V inputs

Channel 2: 24 V present -> D3 LOW, LED lit



ISOLATION



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File: pc817_bench.kicad_sch

Title: PC817 24 V input test bench

Size: A4	Date: 2026-10-02	Rev: 0.4
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